

NGO QUANG MINH

Software Engineer
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Github | LinkedIn | Portfolio
Posts and Telecommunications Institute of Technology (PTIT)

Technical Skills

- **Backend**: **Java 21 (Spring Boot)**, **NestJS**, Python, OOP, RESTful APIs
- **Frontend**: **ReactJS**, TypeScript, Vite, Tailwind CSS, **TanStack Query**, HTML5/CSS3, JavaScript
- **Databases & Caching**: **MySQL**, **PostgreSQL**, **Redis**, Database Modeling & Query Optimization
- **DevOps, Cloud & Infra**: **Docker**, **CI/CD (GitHub Actions)**, **Linux**, VPS Management, AWS (EC2, S3), Nginx Reverse Proxy, Cloudflare Tunnel
- **Git & Version Control**: Git Flow, Branching, Rebase, Cherry-pick, Conflict Resolution, SSH Key-based Authentication
- **Machine Learning & AI**: **YOLO**, **CNN (FaceNet)**, ResNet-50, Claude Code, Gemini API, MCP Integration

Work Experience

- **MEKONG CO., LTD** *01/2026 – 09/2026*
  Full-Stack Developer Intern Trend Coffee | The Terminal
  – Co-developed and maintained production web platforms for an F&B chain (Order, Customer Service, Warehouse), supporting daily business operations.
  – Engineered scalable backend RESTful APIs using **NestJS**, implemented **Redis** caching for high-frequency queries, and integrated **Firebase Cloud Messaging (FCM)** for real-time marketing push notifications.
  – Built responsive frontend interfaces with **ReactJS, TypeScript, and TanStack Query** to strengthen system availability and user experience.
  – Enhanced system security and performance using **Nginx** as a reverse proxy with automated and centralized SSL certificates.
  – Managed deployments on **Linux VPS** environments utilizing **Docker Compose** and automated **GitHub Actions** CI/CD pipelines.
  – Monitored server metrics (CPU, memory, disk usage) and diagnosed system bottlenecks through access and error logs to maintain high uptime.
  – **Impact:** Successfully processed over 9,000+ requests per day with high transaction stability during peak hours.
  – **Result:** Attracted **700+ new active users** within the first 3 months of launch.

Featured Projects

- **MindCare – Psychological Counseling Platform** *01/2026 – 04/2026*
  Microservices, AI & WebRTC Demo | API Docs | Observer
  – Architected a polyglot microservices system (**Java, Go, Python, Node.js**) routed through **Kong API Gateway**, developing a custom Go plugin for centralized JWT validation, RBAC, and strict internal route isolation (`/internal/*`).
  – Implemented **M2M Authentication** using asymmetric RSA tokens and configured asynchronous communication with **RabbitMQ** (Topic Exchanges, Persistent Queues).
  – Built a resilient payment flow using the **Transactional Outbox Pattern** with Exponential Backoff Retries to handle **VNPay** IPN webhooks and synchronize booking states.
  – Integrated AI psychological screening via **Google Gemini API** and created an emotional support chatbot using local embeddings with **Ollama** and **PostgreSQL PGVector** for semantic retrieval.
  – Engineered real-time messaging (**Socket.IO, Redis**) and 1-on-1 peer-to-peer audio calling via **WebRTC**.
  – **DevOps & Infra:** Containerized all 8+ services with multi-stage **Dockerfiles**, orchestrated deployment on a **Linux VPS** via **Docker Compose**, and secured ingress routing using **Cloudflare Tunnel**.

- **Find Anyone Online** *10/2024 – 02/2025*
  Spring Boot & FaceNet Vector Search Github
  – Engineered a facial recognition pipeline utilizing **FaceNet (CNN)** embeddings and high-dimensional vector search.
  – Containerized the deep learning pipeline with **Docker** and integrated it seamlessly with a **Spring Boot** backend and **ReactJS** frontend.

Education & Certifications

- **Posts and Telecommunications Institute of Technology (PTIT)** *2022 – 2027*
  B.E. in Information Technology (GPA: 3.0/4.0)
  – Relevant Coursework: Java Programming (4.0/4.0), C++ Programming (4.0/4.0), Object-Oriented Programming (3.7/4.0).
  – Scientific Research: Apartment Building Fee Management System – Scored 82/100 defense.

- **Cisco Certified Network Associate (CCNA Curriculum)**, Network Topology Design & Cisco IOS Configuration *2025*
- **Methodologies & Languages**, English (TOEIC 600+), Agile/Scrum, Git Flow, CI/CD Automation *2026*